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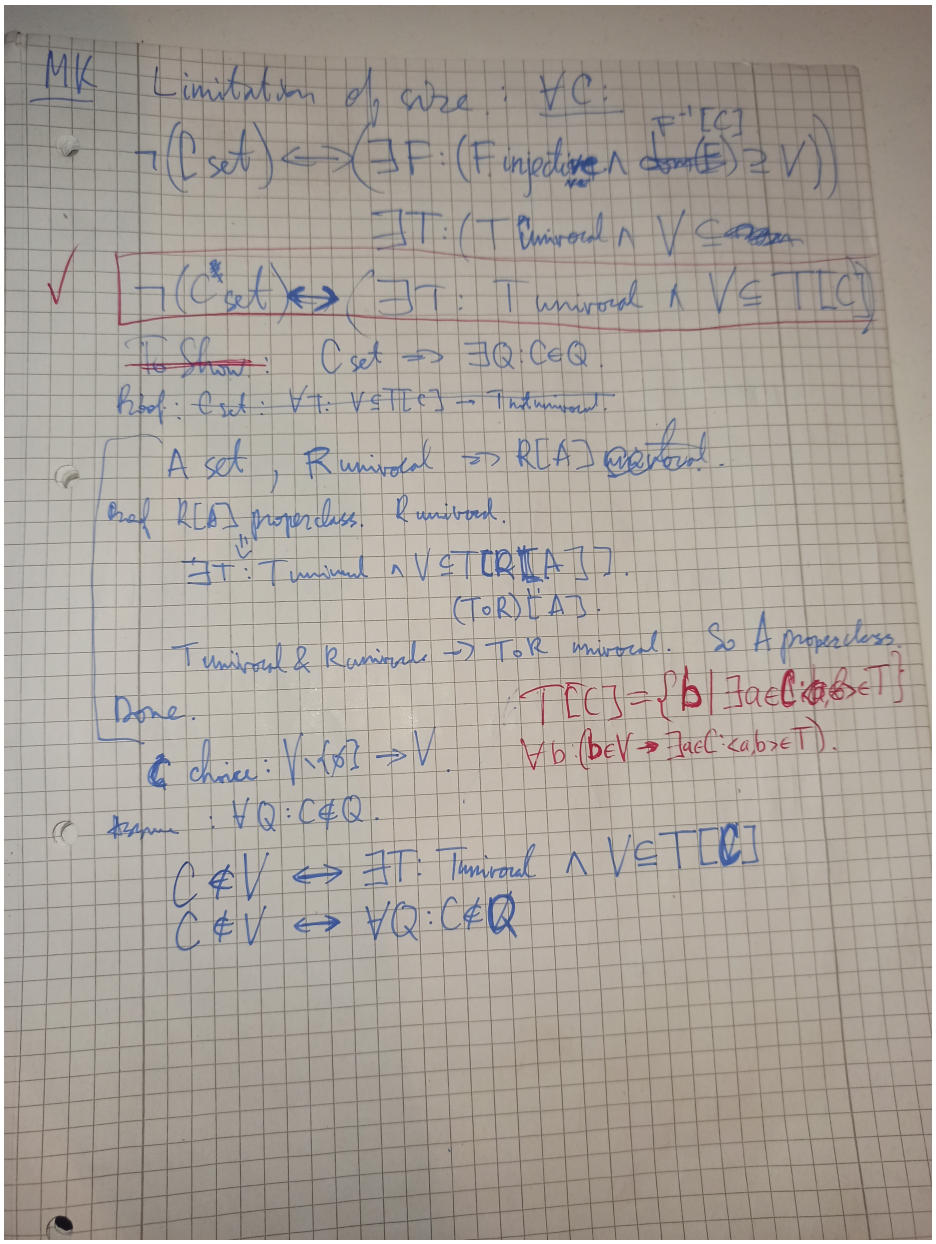
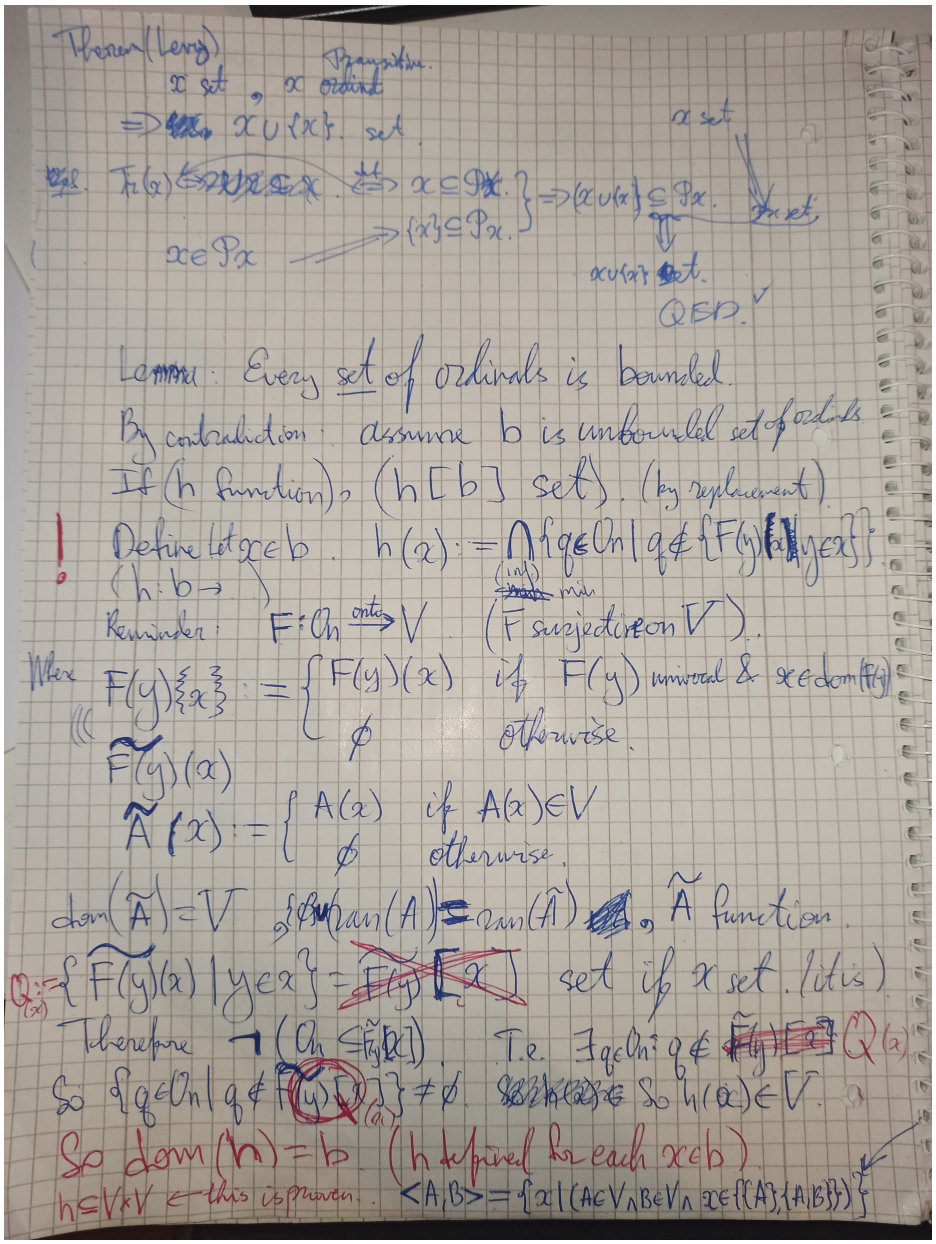
June 24, 2024

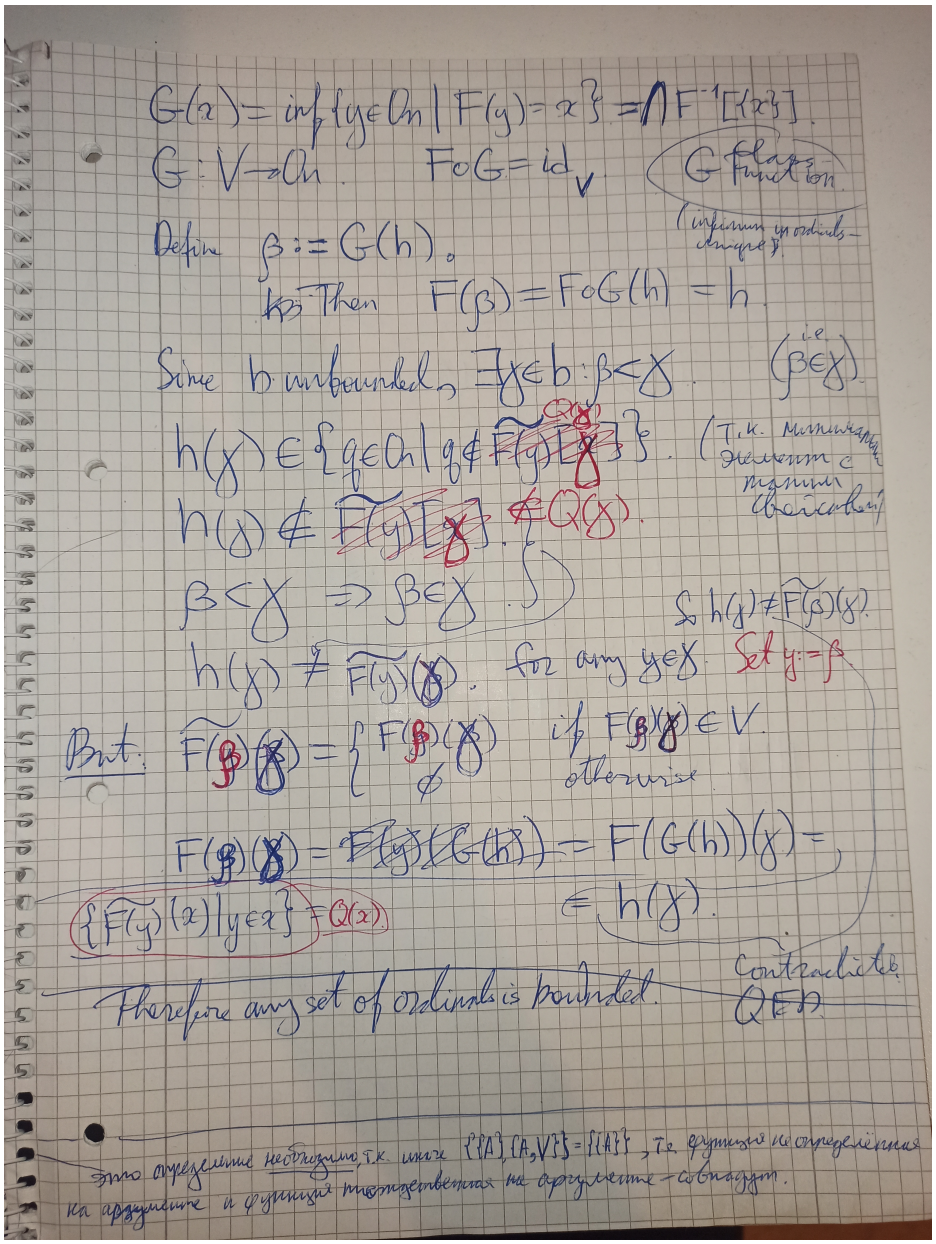
Theorem 0.1. (Azriel Levy 1968 "On Von Neumann's Axiom System for Set Theory" https://www.jstor.org/stable/2315201)

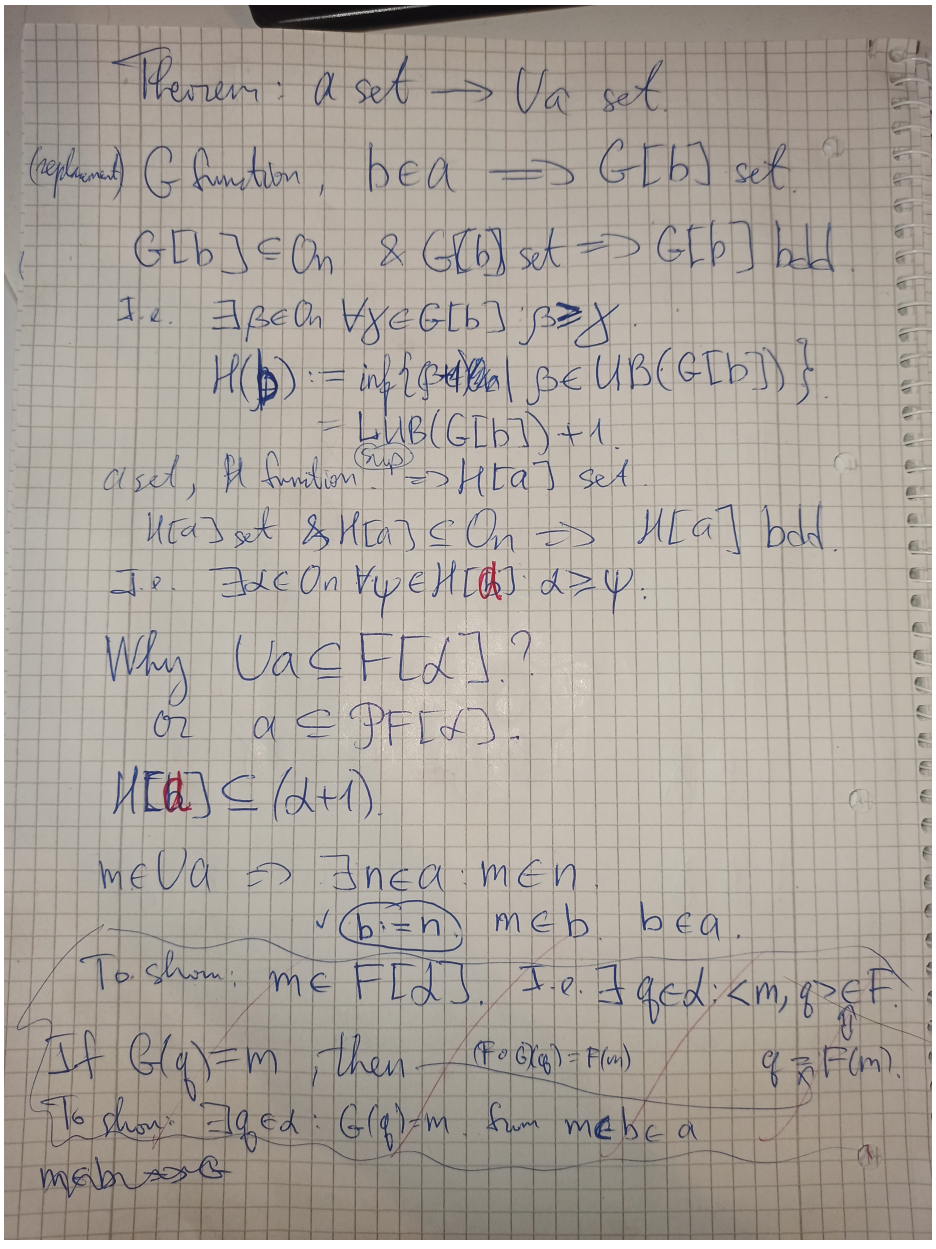
$$(A \text{ set}) \longrightarrow \left(\bigcup A \text{ set}\right)$$

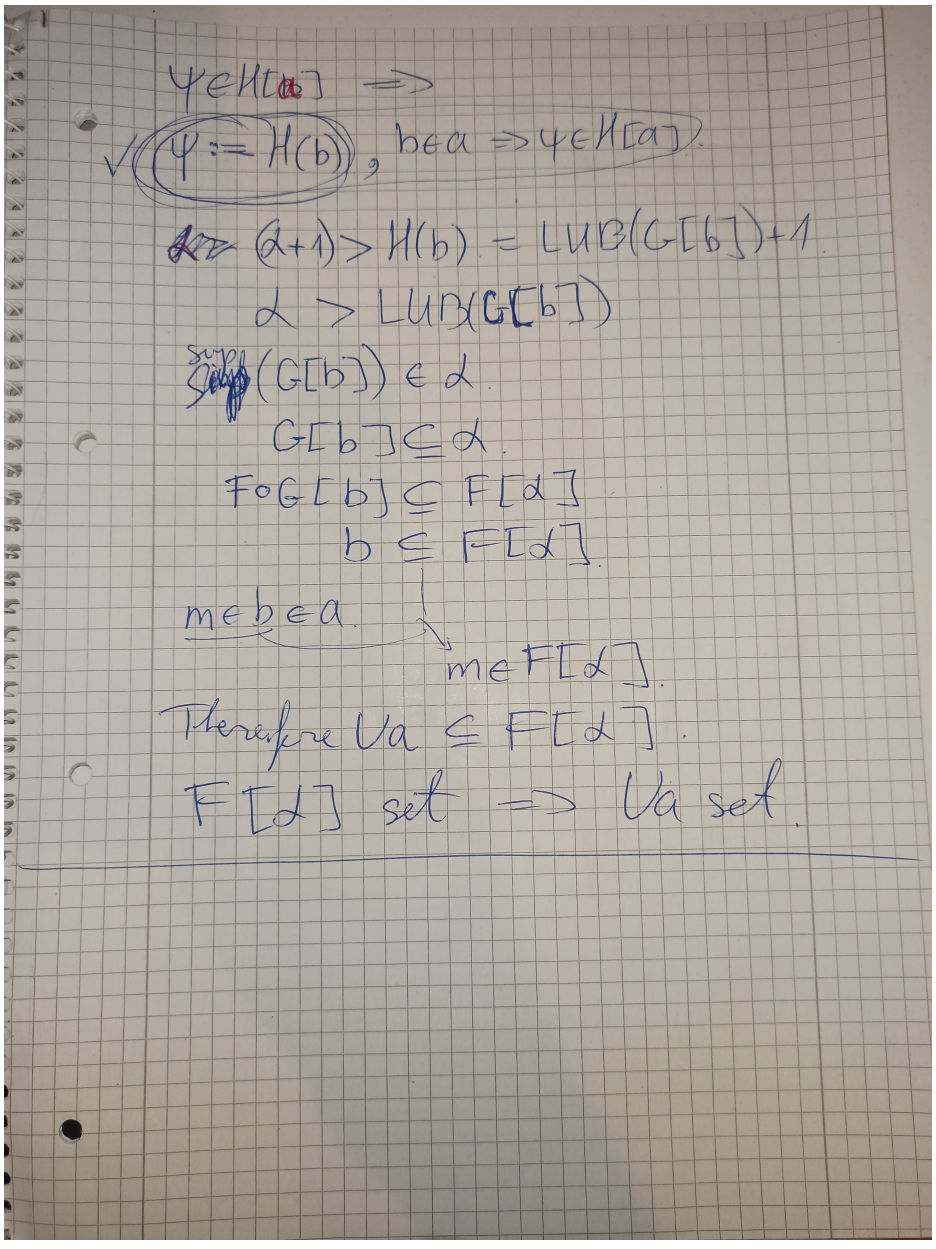
Proof.
(the proof needs theorems about ordinals done first)
...
Done.


MK Limitation of size : ∀C:
¬(C set) ⟷ (∃F: (F injective ∧ F⁻¹[C] ⊇ V))
∃T: (T universal ∧ V ⊆
¬(C set) ⟷ (∃T: T universal ∧ V ⊆ T[C])
C set ⇒ ∃Q: C∈Q.
A set, R universal ⇒ R[A] universal.
R[A] proper class. R universal.
∃T: T universal ∧ V ⊆ T[R[A]].
(T∘R)[A].
T universal & R universal → T∘R universal. So A proper class.
T[C] = {b | ∃a∈C: ⟨a,b⟩∈T}
∀b: (b∈V → ∃a∈C: ⟨a,b⟩∈T).
Done.
choice: V\{0} ⇒ V.
: ∀Q: C∉Q.
C∉V ⟷ ∃T: T universal ∧ V∈T[C]
C∉V ⟷ ∀Q: C∉Q


Theorem (Levy)
x set, x ordinal. Transitive.
⇒ x ∪ {x} set.
x set
x ∈ Px ⇒ {x} ⊆ Px
x ∪ {x} set.
QED.
Lemma: Every set of ordinals is bounded.
By contradiction: assume b is unbounded set of ordinals
If (h function), (h[b] set). (by replacement)
Define let x ∈ b. h(x) := ⋂{q ∈ On | q ∉ {F(y) | y ∈ x}}
(h : b →)
(inf) min
Reminder: F : On ⇒ V (F surjective on V).
Where F(y){x} := { F(y)(x) if F(y) ordered & x ∈ dom(F(y))
∅ otherwise.
F̃(y)(x)
Ã(x) := { A(x) if A(x) ∈ V
∅ otherwise.
dom(Ã) = V ⇒ ran(A) = ran(Ã), Ã function.
Q(x) := {F̃(y)(x) | y ∈ x} = F̃(y)[x] set if x set. (it is)
Therefore ¬(On ≤ F̃(y)[x]) i.e. ∃q ∈ On : q ∉ F̃(y)[x] (Q(x))
So {q ∈ On | q ∉ F̃(y)[x]} ≠ ∅ So h(x) ∈ V.
So dom(h) = b (h defined for each x ∈ b)
h ⊆ V × V ← this is proven. <A,B> = {x | (A ∈ V ∧ B ∈ V ∧ x ∈ {{A},{A,B}})}


$G(\alpha) = \inf\{\gamma \in On \mid F(\gamma) = \alpha\} = \bigcap F^{-1}[\{\alpha\}]$.
$G : V \to On$. $F \circ G = id_V$.
G flags function.
(infimum of ordinals - unique)
Define $\beta := G(h)$.
Then $F(\beta) = F \circ G(h) = h$.
Since h unbounded, $\exists \gamma \in b : \beta < \gamma$. (i.e. $\beta \in \gamma$).
$h(\gamma) \in \{\eta \in On \mid \eta \notin \widetilde{F(\gamma)}[\gamma]\}$. (т.к. минимальный элемент с заданным свойством)
$h(\gamma) \notin \widetilde{F(\gamma)}[\gamma] \in Q(\gamma)$.
$\beta < \gamma \Rightarrow \beta \in \gamma$.
$h(\gamma) \neq \widetilde{F(\gamma)}(\gamma)$. for any $\gamma \in X$. Set $\gamma := \beta$
$\Rightarrow h(\beta) \neq \widetilde{F(\beta)}(\beta)$.
But: $\widetilde{F(\beta)}(\beta) = \begin{cases} F(\beta)(\beta) & \text{if } F(\beta)(\beta) \in V \\ \emptyset & \text{otherwise} \end{cases}$
$F(\beta)(\beta) = F(\gamma)(G(h)) = F(G(h))(\gamma) =$
$= h(\gamma)$.
$\{\widetilde{F(\gamma)}(x) \mid \gamma \in x\} = Q(x)$
Therefore any set of ordinals is bounded.
Contradiction.
QED.
это определение необходимо т.к. иначе $\{\{A\}, \{A, V\}\} = \{\{A\}\}$, те функция по определению на аргументе и функция тождественная на аргументе - совпадут.


Theorem: a set → ⋃a set.
(replacement) G function, b∈a ⇒ G[b] set.
G[b] ∈ On & G[b] set ⇒ G[b] bdd.
i.e. ∃β∈On ∀γ∈G[b]: β≥γ.
H(b) := inf{β∈On | β∈UB(G[b])}
= ⊔UB(G[b]) + 1.
a set, H function (sup) ⇒ H[a] set.
H[a] set & H[a] ⊆ On ⇒ H[a] bdd.
i.e. ∃α∈On ∀ψ∈H[a]: α≥ψ.
Why ⋃a ⊆ F[α]?
or a ⊆ PF[α].
H[a] ⊆ (α+1).
m∈⋃a ⇒ ∃n∈a: m∈n.
b:=n. m∈b. b∈a.
To show: m∈F[α]. i.e. ∃q∈α: <m,q>∈F.
If G(q)=m, then (F∘G)(q) = F(m). q ≥ F(m).
To show: ∃q∈α: G(q)=m. from m∈b∈a


ψ∈H[α] ⇒
ψ := H(b), b∈α ⇒ ψ∈H[α].
(α+1) > H(b) = LUB(G[b])+1
α > LUB(G[b])
sup(G[b]) ∈ α.
G[b] ⊆ α.
F∘G[b] ⊆ F[α]
b ⊆ F[α].
m∈b∈α.
m∈F[α].
Therefore Vα ⊆ F[α].
F[α] set ⇒ Vα set.